Genomics of Gene Expression Lab

AI models to interpret system biology models

PaintOmics AI, powered by TELLME: does the AI read the data the way its paper did?

Tianyuan Liu · Ana Conesa
I²SysBio (CSIC–Universitat de València)
Chair: Ana Rojas

SEBiBC 2026 · Toledo · Friday 13 November, 10:00–11:00

paintomics.org

Slides and materials

Pick a paper: six published datasets, already run on paintomics.org. They are on slide 27.

Then mark: in the paper's form, give each statement the AI wrote one of four marks.

Each paper has its own form: its QR code and a short link are on slide 27, on the paper's card.

elixir

CSIC
Consejo Superior de Investigaciones Científicas

i²sysbio INSTITUTE FOR INTEGRATIVE SYSTEMS BIOLOGY

The hour

Genomics of Gene Expression Lab

1 Why, and how it works — from 10:00

- One table, many stories: why reading a pathway analysis is the hard part
- PaintOmics: every omic painted on its pathways
- TELLME: agents that walk the pathways and write; what it can and cannot do

2 Your turn — until 10:47

- Pick one of six published papers
- Read what the AI wrote about that paper's data
- Give every statement one of four marks

3 What the room found — to 11:00

- Your marks, live, statement by statement
- Next to the marks of a second AI reader
- Your examples

By 11:00 you will have read, and judged, a real AI interpretation yourself.

01

The problem

One table, many stories

Genomics of Gene Expression Lab

Pathways summary

STATegra — real mouse Ikaros time course (5 omics)

1,017
Pathways found

114
Significant

Which story do you tell?

- Five omics, six time points
- A paper tells one story

Pathway enrichment

Paint		Pathway name	Features		Significance tests						
			Unique genes	Unique metabol.	DNase-seq	Gene expression	Proteomics	miRNA-seq	Metabolomics	Combined pValue (Fisher)	Reactome Class pValue (Fisher)
☑	K	Cytokine-cytokine receptor interaction	186	0	0.07746	1.5038e-8	-	0.77603	-	2.1573e-7	-
☑	K	Cholinergic synapse	98	0	5.1473e-7	0.03830	1.00000	0.66581	-	1.5489e-5	-
☑	K	Morphine addiction	74	2	1.5899e-5	5.9859e-4	1.00000	0.64622	0.91429	3.8062e-5	-
☑	K	Viral protein interaction with cytokin…	57	0	0.45714	2.6621e-6	-	0.43972	-	6.4059e-5	-
☑	K	Parathyroid hormone synthesis, secr…	103	0	1.1675e-5	0.29583	1.00000	0.02602	-	7.7345e-5	-
☑	R	GPVI-mediated activation cascade	42	0	8.3788e-5	0.18456	0.45916	0.01381	-	8.3140e-5	4.8303e-6
☑	R	Cell surface interactions at the vascul…	156	0	0.02393	1.2567e-4	0.35795	0.30578	-	2.2432e-4	4.8303e-6
☑	K	Ribosome biogenesis in eukaryotes	76	0	0.95200	4.3279e-6	0.20370	0.39801	-	2.2706e-4	-
☑	K	Aldosterone synthesis and secretion	91	1	4.7756e-5	0.01727	1.00000	0.39801	0.70000	6.9025e-4	-

Problem 1: we read what we already know

1 Too much to read



In 148 transcriptomics studies, fewer than 1 hit gene in 100 reached the title or abstract.

Richardson et al., eLife 2024

2 Familiar first

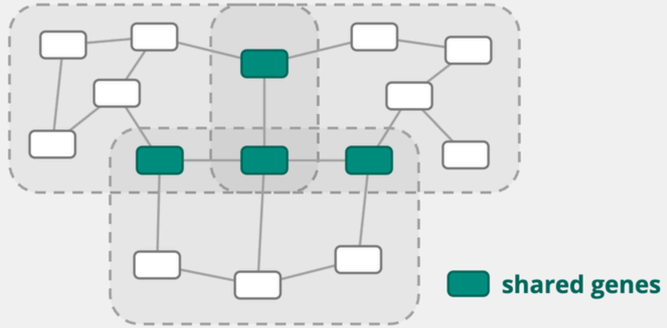


What is published about a gene follows how much is already known about it, not how strongly it changes in the data.

Stoeger et al., PLoS Biol 2018

Haynes et al., Sci Rep 2018

3 One pathway at a time



Pathways share genes, so one pathway's p-value moves with the others, and one large KEGG map can mix up to 21 processes.

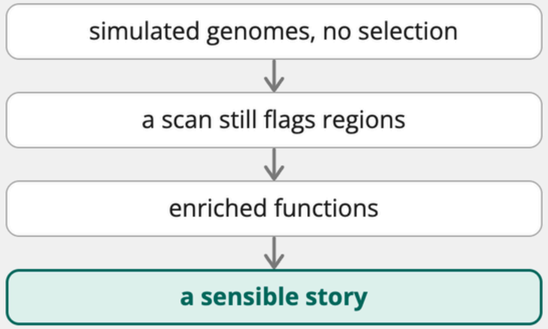
Donato et al., Genome Res 2013

Karp et al., BMC Genomics 2021

The bottleneck is not the statistics: it is one person reading selectively.

Problem 2: a story is easy to build and hard to repeat

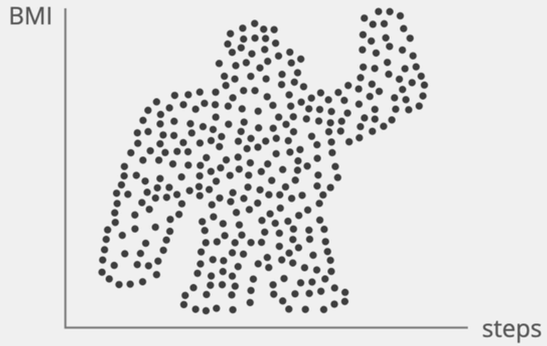
1 A story from any list



Genomes simulated with no selection still gave enriched gene functions and a sensible biological narrative.

Pavlidis et al., Mol Biol Evol 2012

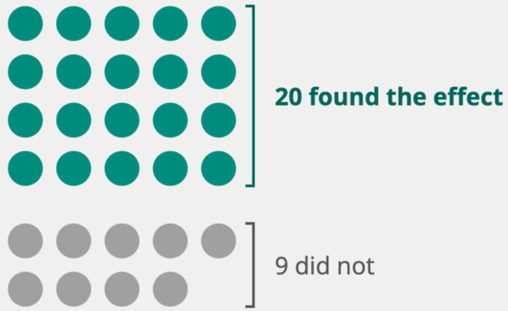
2 We find what we look for



33 students analysed this dataset. Those asked only what they could conclude were almost five times more likely to see the gorilla than those given hypotheses to test.

Yanai & Lercher, Genome Biol 2020

3 Same data, different answers



29 teams, one dataset, one question. And of 235 published enrichment analyses, 35 had no major flaw in statistics or reporting.

Silberzahn et al., Adv Methods Pract Psychol Sci 2018

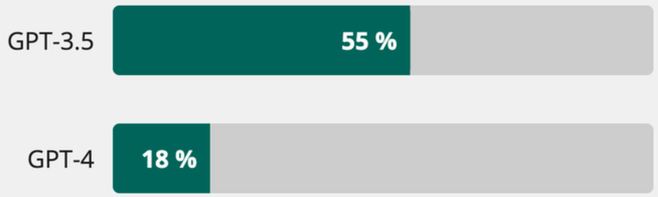
Wijesooriya et al., PLoS Comput Biol 2022

A plausible story is not evidence: it needs a test, and a record of what was read.

Problem 3: asking a language model is not enough

Genomics of Gene Expression Lab

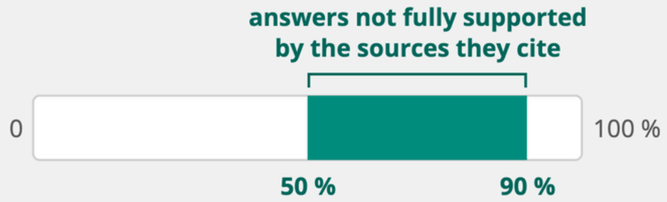
1 References that do not exist



Of 636 references in literature reviews written by ChatGPT, this share was invented.

Walters & Wilder, Sci Rep 2023

2 Sources that do not support the sentence



Seven models, 800 medical questions. Even GPT-4o with web search left about 30 % of statements unsupported.

Wu et al., Nat Commun 2025

3 Gene functions from memory



GPT-4 naming Gene Ontology gene sets. Checking each claim against databases is what cuts invented functions.

Hu et al., Nat Methods 2025

Wang et al., Nat Methods 2025

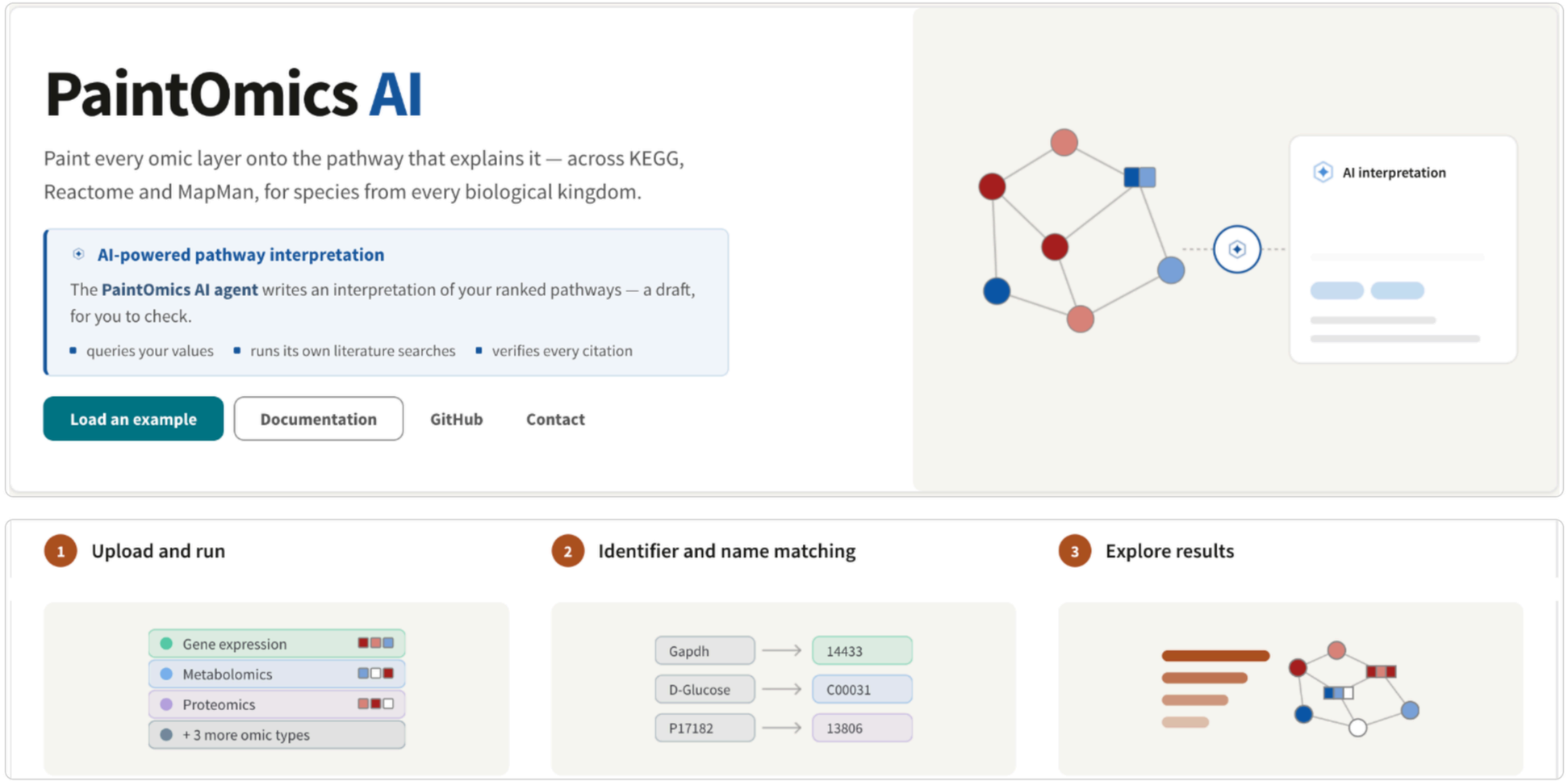
Measured on the models of 2023 to 2025.

Fluent is not checked: every sentence has to rest on something that was read.

02

Our solution: PaintOmics and TELLME

PaintOmics in one picture

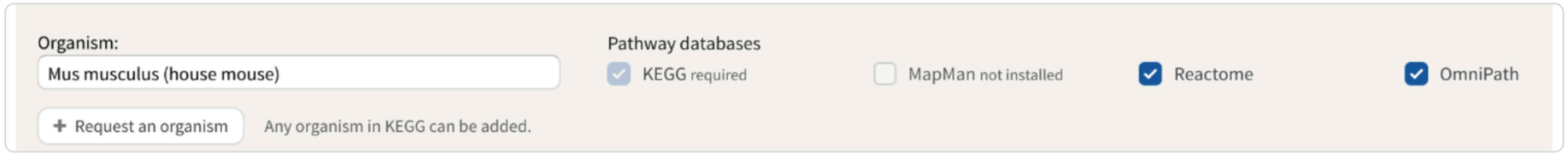


Paint every omic layer onto the pathways that explain it: KEGG, Reactome, MapMan and OmniPath, for any KEGG organism.

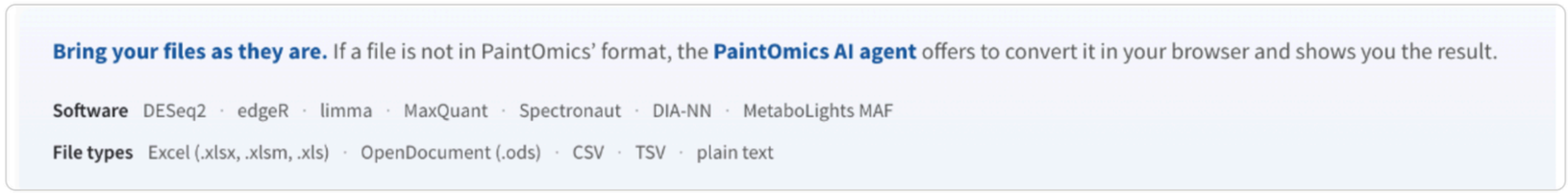
1 · Upload and match

Genomics of Gene Expression Lab

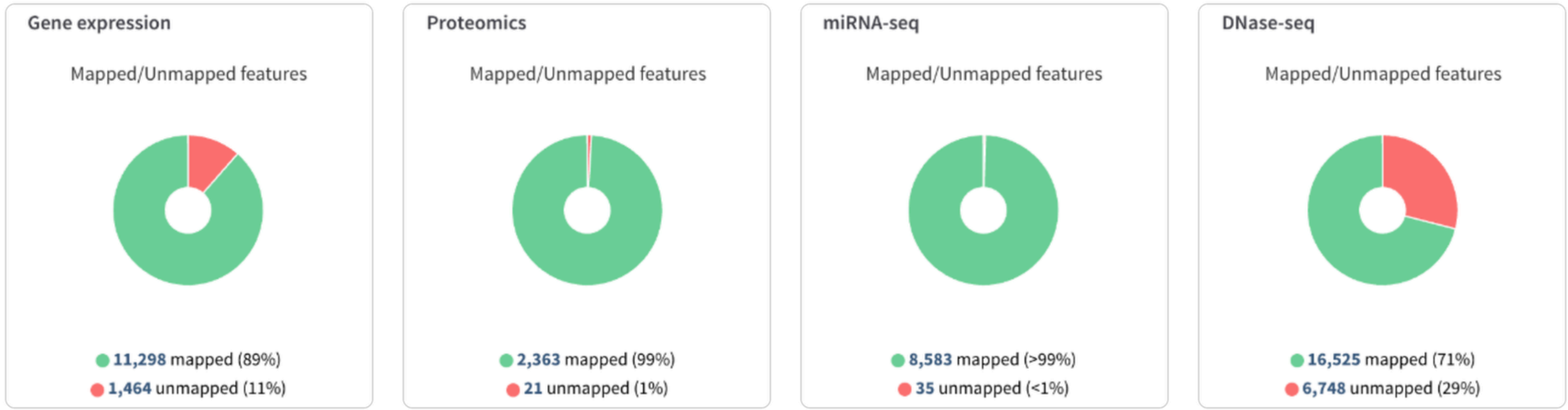
choose organism and databases



your files as they are



IDs matched, per omic



2 · Which pathways change

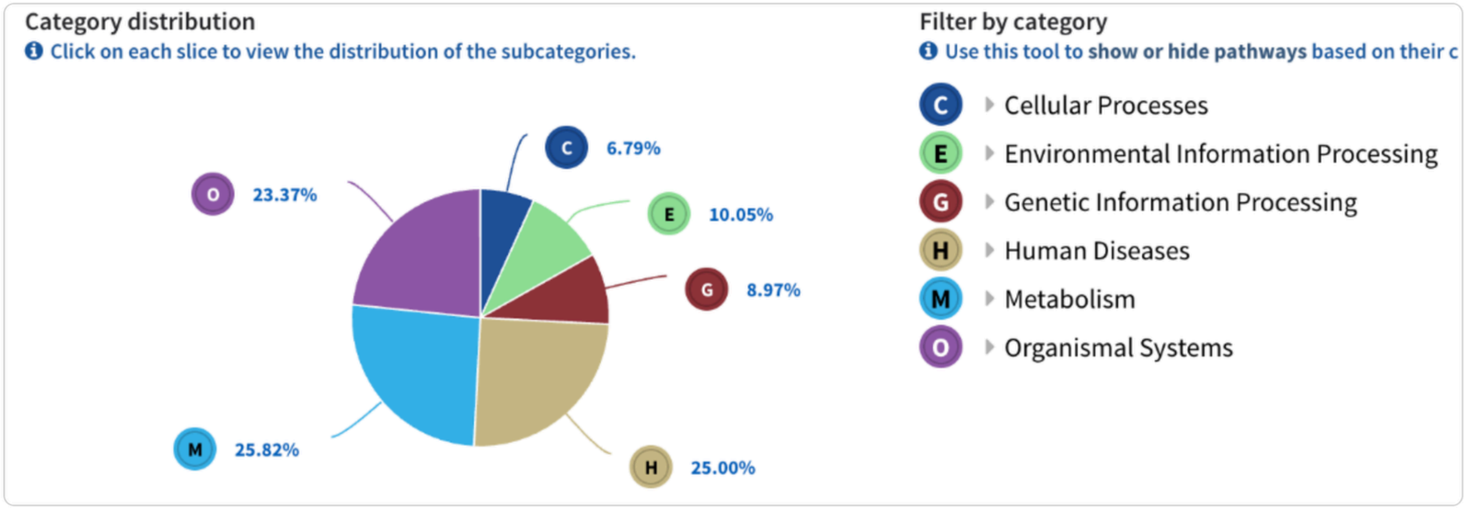
Genomics of Gene Expression Lab

significant per database

	DATABASE	FOUND	SIGNIFICANT
K	KEGG	367	73
O	OmniPath	120	10
R	Reactome	530	31

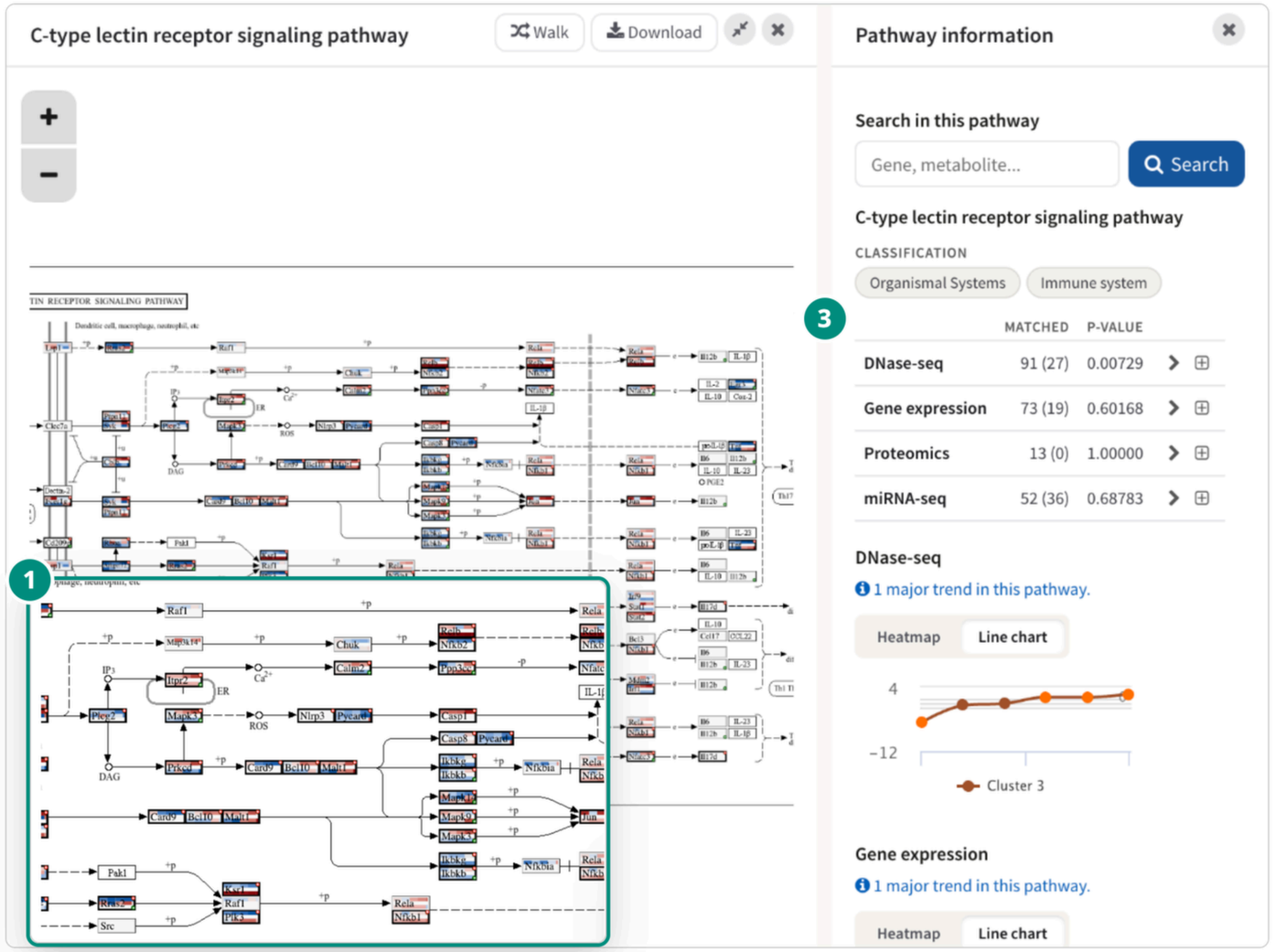
One explorer tab per database below; the enrichment table lists all of them, filterable from its search bar.

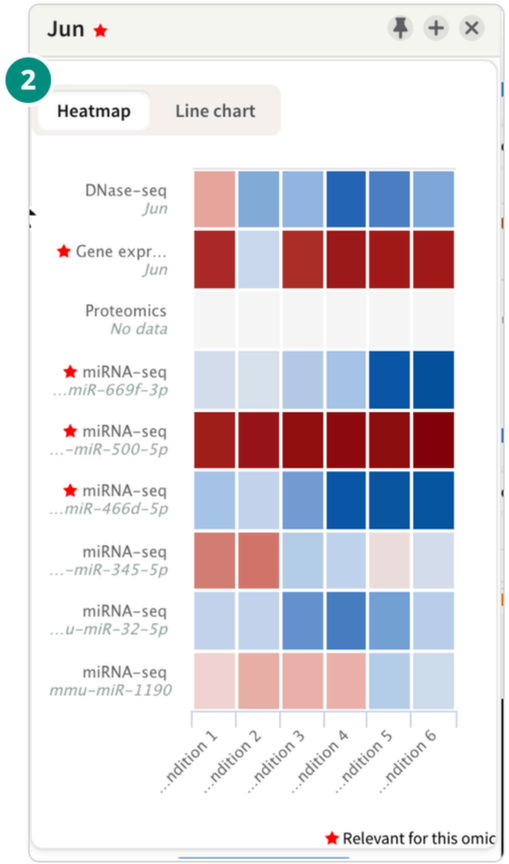
by category



a p-value per omic, and a combined one: the enrichment table on One table, many stories

3 · Paint every omic on the pathway



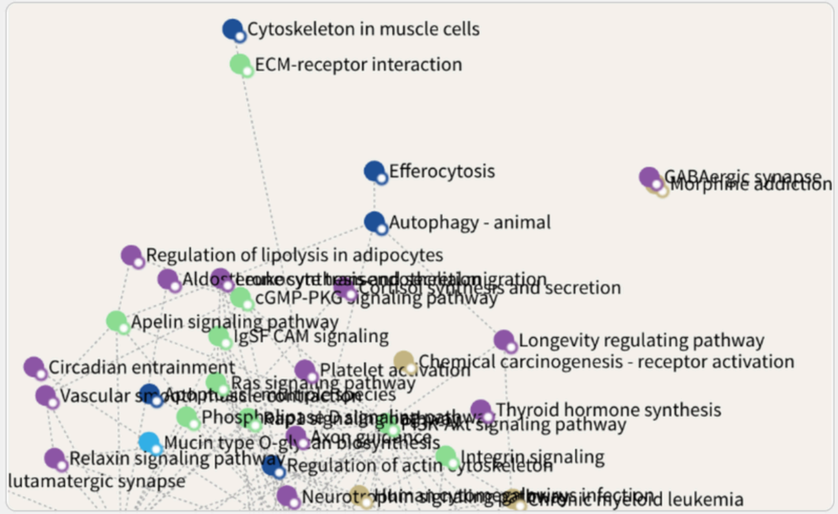


1. each gene box: your omics × your time points
2. one gene, all layers (★ = relevant)
3. per-omic p-value and trend for the pathway

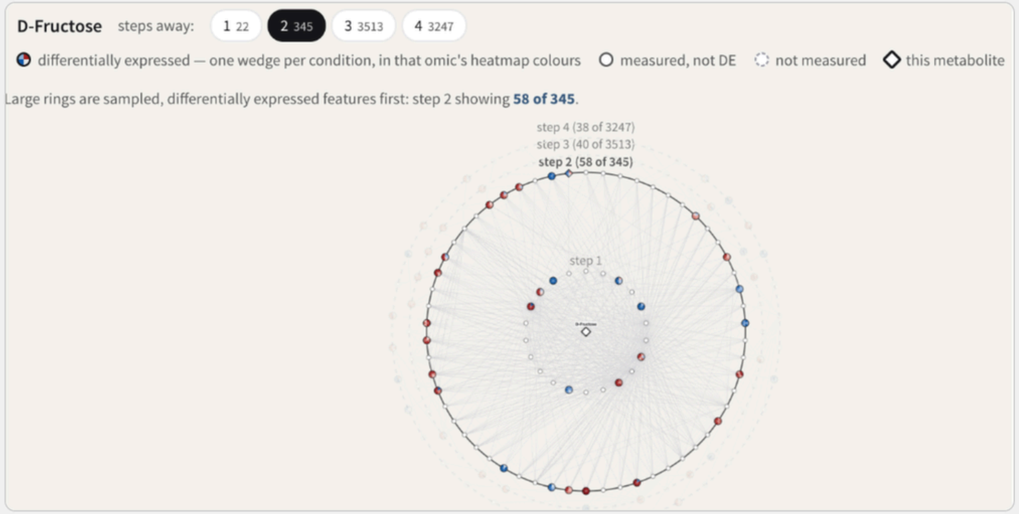
Beyond single pathways

Genomics of Gene Expression Lab

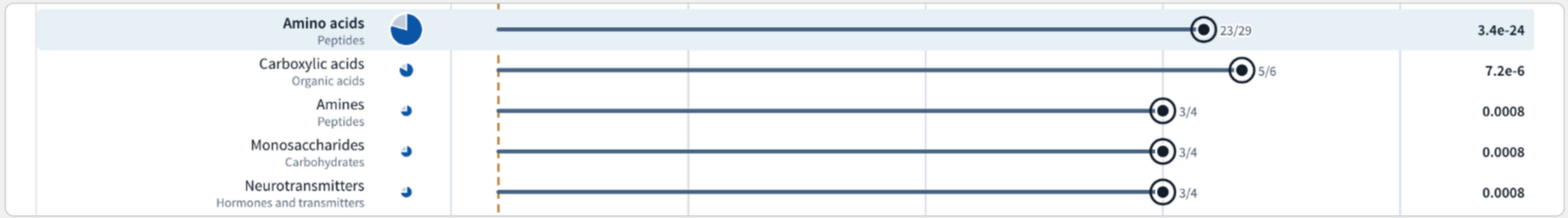
pathways linked by shared biology



metabolites with changed genes around them



metabolite classes that respond



PaintOmics shows everything. Reading it is still up to you.

Read every pathway, not the familiar ones

By hand

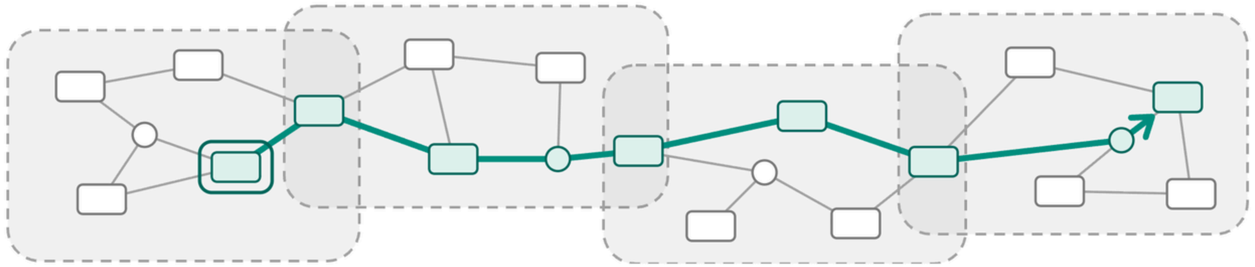


the ones we know

one pathway at a time, the familiar ones first

TELLME

pathway border — the walk



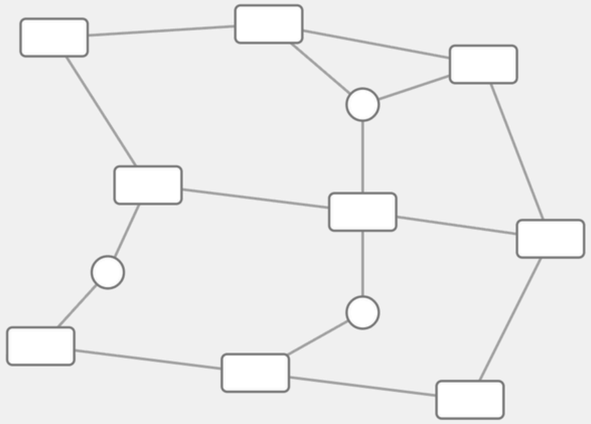
1. **Read:** every KEGG, Reactome and OmniPath pathway for your organism, joined into one network with your values on it
2. **Walk:** start where your relevant genes cluster and follow interactions across pathway borders
3. **Write:** every statement cites a leg or a paper, checked the same way every time

Same rules for every pathway: a systematic reading, not a familiar story.

TELLME in one picture

Genomics of Gene Expression Lab

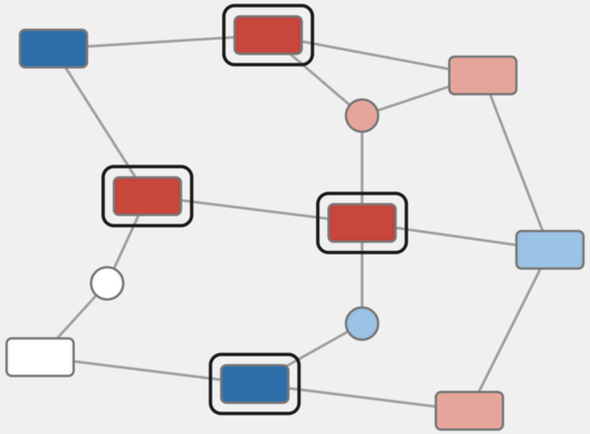
1 Knowledge graph



KEGG · Reactome · OmniPath

gene compound

2 Your values on every node



your values, your relevant list

down up relevant

3 Agents walk it and write

Results

Ikzf1 → Gata3 [1]

Gata3 → Tbx21 [2]

From the graph and your values to a text with citations.

Where it runs

Genomics of Gene Expression Lab

Your browser

PaintOmics server

Supercomputador Drago (CSIC)

job queue

the walk runs here, beside the analyses

agents

they walk, write and check: the next slide

LLM gateway

`llm.iiia.es`

IIIA-CSIC, the Artificial Intelligence Research Institute of CSIC

hardware in Spain

OpenAI-compatible API

Pace: the key allows 60 requests/min; every caller is paced at 54/min

Model

`DeepSeek-V4-Flash-0731`

a pinned, dated snapshot, for reproducibility

fallback: the gateway's alias **`default/llm`**, only if the pinned model is not served

every result records the model that answered

Literature

NCBI PubMed

search, best-match order

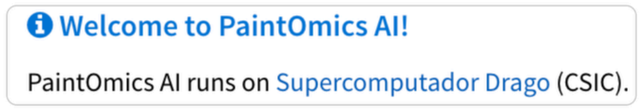
Europe PMC / PubMed Central

full text (introduction, results, discussion) when available; otherwise the abstract

What leaves the server

feature names, the values and column labels of the nodes walked or shown, and the experiment description

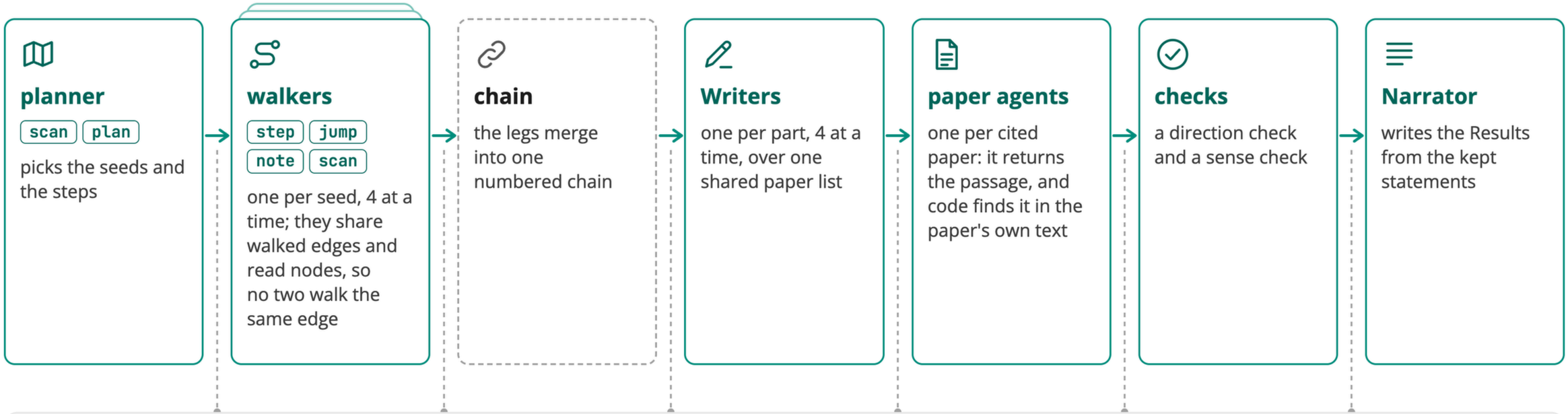
Never: your files, unmapped features or account data



the welcome dialog of paintomics.org

The agents and their tools

Genomics of Gene Expression Lab



Between every step: code guards

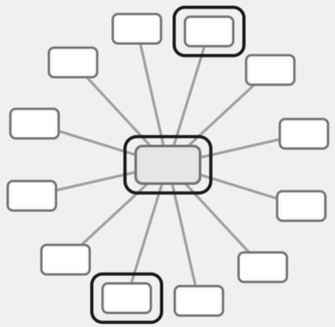
Refused: a step to a non-neighbour · the same edge twice · a reading that names no data layer · a value not in your data

The budgets of each kind of walk (seeds, statements, papers, words, time): Pathway walk vs network walk.

Picking the seeds

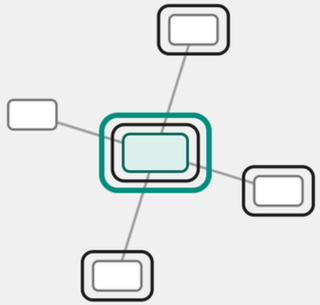
Heat

relevant



a hub:
many neighbours, few relevant

cool



a small node:
most neighbours relevant

hot

how surprising the number of relevant features among a node's neighbours is, given how many neighbours it has (a hypergeometric test)

So a hub gene is not hot just for being a hub.

1 Seed candidates

relevant nodes that are not next to a hotter candidate

2 Anchored to the perturbation

the design card names the perturbed gene, and its known transcription-factor targets (CollecTRI, TRRUST) are added as edges

seeds come first from within two steps of it, hottest first, and the first seed must be among the nearest

STATegra: the network walk was anchored on Ikzf1 (its anchor line: What you will see)

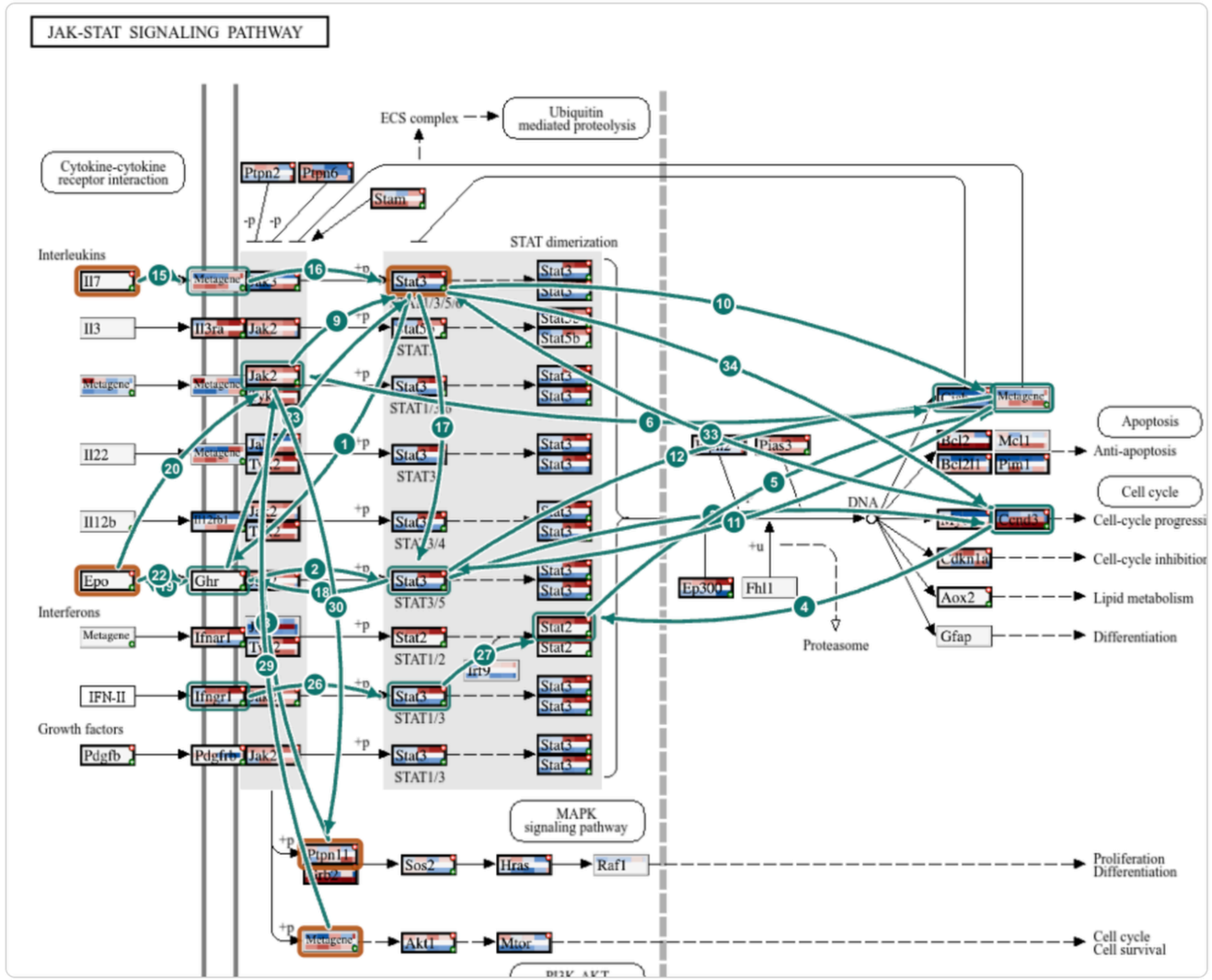
3 What a node is

genes, compounds (metabolites) and your miRNAs

currency metabolites (ATP, ADP, phosphate, NAD, water…) are never walked and never a seed: no walk joins two pathways through water

Reactome complexes are read as their member proteins; a reaction that would join more than 36 pairs is skipped

How the walk moves



the planner's plan

PLAN Stat3, Ptpn11, Il7, Epo, Pik3r1 · 30 steps

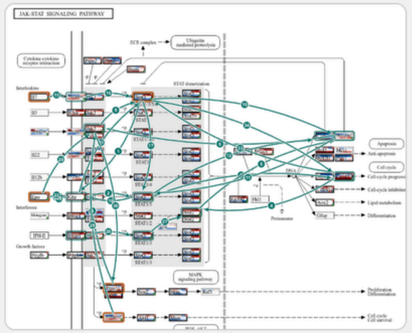
The perturbation is Ikzf1 (Ikaros), an induced transcription factor, so the walk should start where the perturbation acts and move outward. Stat3 is the hottest candidate (heat 2.25, 36/77 relevant) and sits at the hub of the JAK-STAT arm of this pathway; Ptpn11 (SHP2, 8/11 relevant) is the

1. Start at a seed of the plan
2. Step only along an interaction a database draws
3. At every stop, read the values against the design

Left: a real walk of the STATegra job on one KEGG map (JAK-STAT). Teal arcs: the legs, in walk order; ringed: the seeds.

Pathway walk vs network walk

Pathway walk

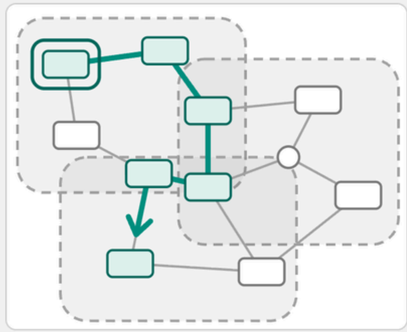


one map, started from the **Walk** column of a Step 4 diagram

edges carry that map's own signs: NF1 → HRAS is an activation on the Ras map, an inhibition on the MAPK map

seeds	up to 5, 3–6 steps each
statements	2–4 per part, up to 4 parts
papers	about 12
Results	400–1,200 words
time	8 minutes
graph-artifact check	advisory: amber, hides nothing

Network walk



the job's interpretation: it starts by itself when Step 2 finishes

one network per organism: every KEGG, Reactome and OmniPath pathway, joined through their shared genes and compounds, so a walk crosses from one pathway into another

on mouse, about 131,000 edges once the job's miRNAs are added

seeds	up to 10, 5–8 steps each
statements	3–5 per part, up to 6 parts
papers	about 24
Results	800–2,000 words
time	10 minutes
graph-artifact check	a failure hides the text

Is the walk a graph artifact?

1. After the walk, a scripted walker with no model walks your real relevant flags once
2. then 50 times with the flags shuffled over the measured nodes
3. and counts modules: stretches of the chain with at least three relevant nodes whose signed arrows agree with the values at least half the time

Also reported: the edges joining two relevant nodes, against chance. A leg through a currency metabolite is an artifact outright.

Pass



more modules on your flags than on shuffled ones (empirical $p < 0.05$), or, when the graph holds only one or two modules, seeds far hotter than chance

Network walk, STATegra

✓ Not a graph artifact

passed: the text is shown

on a network walk, a failure hides the text

each check is a chip above the Results: its numbers on hover, its reason when it fails

Pathway walk, JAK-STAT

! Not a graph artifact

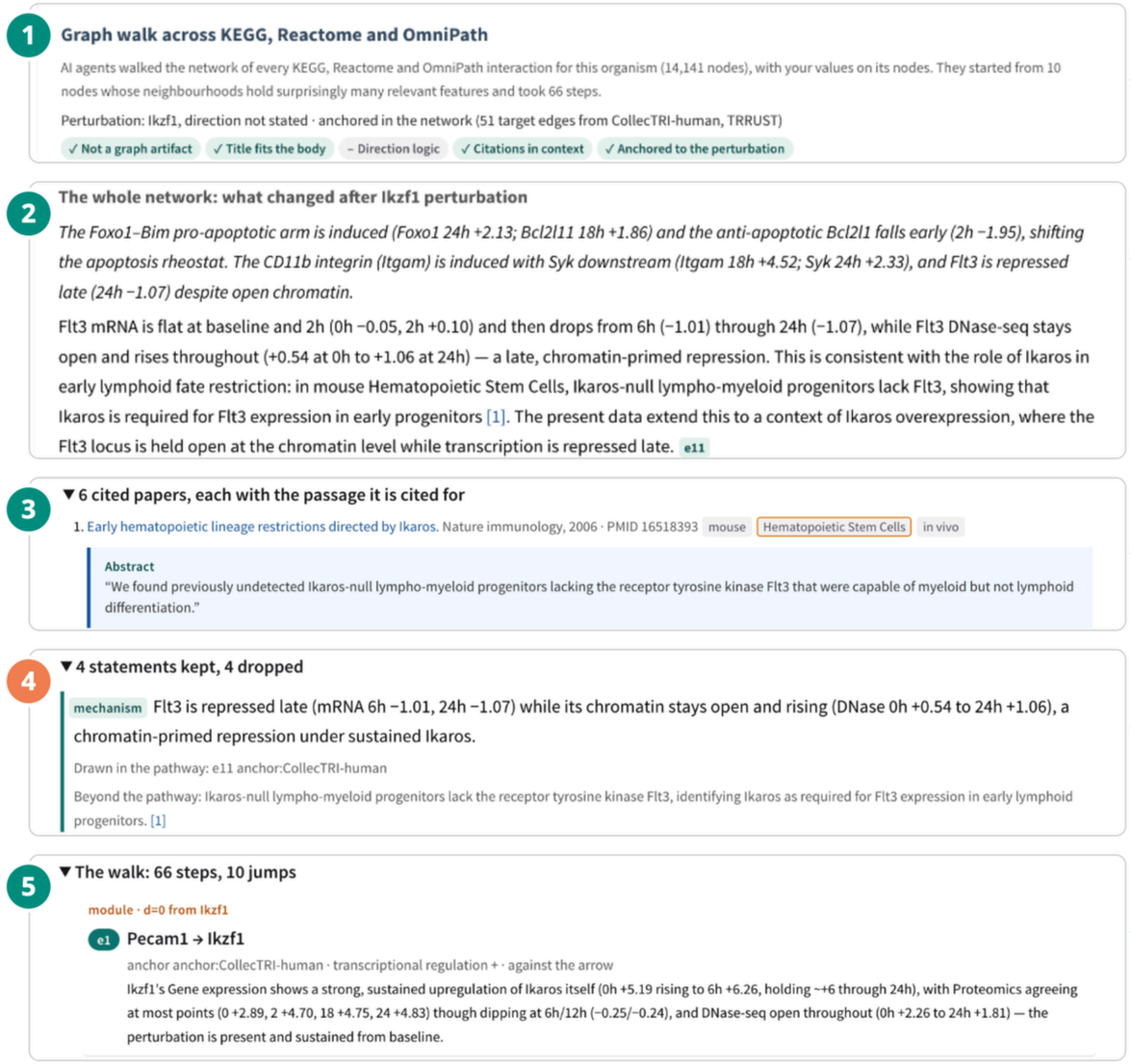
Not a graph artifact (advisory on a pathway walk, hides nothing): the check's walker, which steps only from relevant feature to relevant feature, found 2 modules on the data and 1.6 on shuffled (permuted) data on average ($p = 0.57$), with seeds no hotter than on shuffled data ($p = 0.08$): it could not tell this data from shuffled data on this graph. Of 262 measured features, 91 are relevant; 226 pairs of them are adjacent, 126.9 expected by chance ($p = 0.02$)

advisory: shown amber, it hides nothing

on one map there are often too few relevant genes for the test to decide

What you will see

Genomics of Gene Expression Lab



1. header and checks
   - not a graph artifact
   - the title fits the body
   - direction logic
   - citations in context
   - anchored to the perturbation
2. Results, with [n]
3. cited papers with their passages
4. statements, marked mechanism or association, and the dropped ones with reasons
   You mark these, in order
5. the walk, leg by leg

What TELLME can and cannot do

Genomics of Gene Expression Lab

Can

- read every pathway and walk across them
- quote only your values
- cite only papers it read, with the passage
- check every claim with code
- the same rules every time
- a record of every step
- minutes, not weeks

Cannot

- know your hypothesis unless you write it
- see interactions no database draws, or features that failed to map
- judge whether the argument holds: it can write a fluent paragraph that is wrong about your biology
- read full text behind a paywall: it reads the abstract
- tell a real discovery from a plausible one: there is no answer key

TELLME writes the draft; the judgement is yours. Next, you try it on a published dataset.

03

Practice

How you mark: four marks

Genomics of Gene Expression Lab

Makes sense

Consistent with the data and with what you know. You would accept it from a colleague. It may be well known.

Does not make sense

Something in it is wrong or empty. One wrong part is enough.

Especially good

Correct, and it adds something: a link the paper did not make, or a clearer mechanism.

Can't judge

You cannot tell without more work. Better than a guess.

What can be wrong

things to check before you choose “does not make sense”

Data
a value, a direction or a significance is not what the data shows

Citation
the cited paper does not say this, or is about something else

Biology
wrong as biology, against the consensus

Design
misreads the experiment: the contrast, the baseline, the time point

Generic
true, but says nothing about this experiment

Judge each statement against the paper and what you know. One mark per statement.

Your turn

until 10:47

Genomics of Gene Expression Lab

One form per paper

the bottom QR on the paper's card · no account

A	Hernandez 2017	8 statements
B	Wang 2019	8 statements
C	Jiménez-Munguía 2021	16 statements
D	Balzano-Nogueira 2021	14 statements
E	Rubio 2021	10 statements
F	Fàbregas 2018	13 statements

slides: **github.com/ConesaLab/SEBiBC2026**

1. **Pick a paper**
   six papers, A to F: slide 27
2. **Open its job page and read what the AI wrote**
   the top QR on the paper's card; AI Interpret is top right, with the evidence behind each statement
3. **Open the paper's form**
   the bottom QR on its card, or its short link; every statement the AI wrote is there in full
4. **Give each statement one mark, then press Submit**
   makes sense, does not make sense, especially good, can't judge. None is required: skip what you do not reach

- Your marks count only after Submit: press it by 10:45
- After Submit you can still edit your answers
- Time up? Stop, rather than skip around

Pick a paper

top QR: the job, with the walk and the cited passages · **bottom QR: the form where you mark**

Genomics of Gene Expression Lab

A Hernandez 2017

Does deleting the C5a receptor change microglia in an Alzheimer's model?

Mouse · gene expression, 4 ages × 3 contrasts

Mark it here

`forms.gle/1uJ7VTYMQmAF5Gch6`

Job **`7NHQTK7rWz`** doi.org/10.1186/s13024-017-0210-z

B Wang 2019 iScience

What does HNF4α need to make hepatocytes? Two edited mutants

Human · gene expression + proteomics, 2 mutants vs wild type

Mark it here

`forms.gle/MviRmxDgUZCv99X68`

Job **`Y6n7275a0Z`** doi.org/10.1016/j.isci.2019.05.028

C Jiménez-Munguía 2021

How do brain endothelial cells respond to pneumococcus and to its adhesin alone?

Human · gene expression, 2 treatments vs control

Mark it here

`forms.gle/eYt8S9iXbV54zDgn7`

Job **`2Dh754YEIO`** doi.org/10.1038/s41598-021-87021-4

D Balzano-Nogueira 2021

What changes in blood in the year before islet autoimmunity?

Human · gene expression + metabolomics, 5 time windows

Mark it here

`forms.gle/9u6CxLaz8MDkThmKA`

Job **`560S30473C`** held back: “Show the interpretation anyway”

E Rubio 2021 Sci Rep

Minimal hepatic encephalopathy in cirrhosis: what changes in blood?

Human · gene expression + metabolomics + serum cytokines

Mark it here

`forms.gle/RX22xEuCdxguFXbD9`

Job **`3oP1p37B25`** held back: “Show the interpretation anyway”

F Fàbregas 2018

Why do plants that overexpress the BRL3 receptor survive drought?

Arabidopsis · gene expression + metabolomics, genotype × drought

Mark it here

`forms.gle/QyTJ68AqeFpiYCRd6`

Job **`044DBhQFC7`** held back: “Show the interpretation anyway”

Every one of these papers cites PaintOmics and used it on this data. We ran the same data again, with the AI.

What the room found

Genomics of Gene Expression Lab

01 **Each statement**
how many of you chose each mark

02 **Agreement**
where the room agrees, and where it splits

03 **The doubtful ones**
the statements most of you marked “does not make sense”

Live from the six forms

every answer lands in one sheet as it is submitted

Tab “Results”

every statement with its four counts,
a coloured bar of the room's marks,
and the second reader's mark beside it

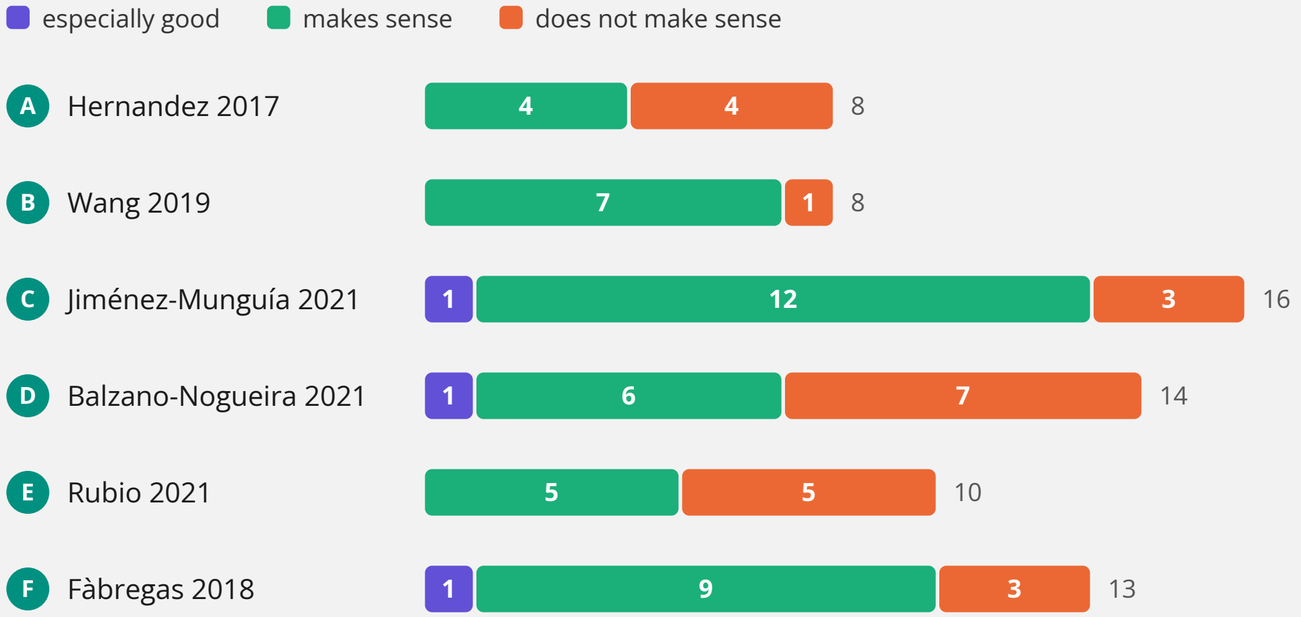
shown from the presenter's laptop

A second reader: an AI marks the AI

Genomics of Gene Expression Lab

1. **Claude Fable 5.2**
   run at its extra-high effort setting, one pass per paper
2. **The same 69 statements**
   and the same four marks you used
3. **With the evidence open**
   the paper's full text, the uploaded values, every cited passage
4. **One more reader**
   an opinion to compare with yours, not an answer key

What it marked, paper by paper



Of 69 statements: **43 make sense · 3 especially good · 23 do not make sense**

Where do you and the second reader disagree? The Results tab shows both.

What the second reader flagged

Genomics of Gene Expression Lab

Data — D · Balzano-Nogueira 2021 · S14

The AI wrote: A dip in APOC2 means less triglyceride clearance.

The value belongs to APC2, another gene, matched to APOC2 through an old alias. APOC2 itself is not in the signature.

Design — A · Hernandez 2017 · S7

The AI wrote: Cx3cr1 is partially restored by the C5aR1 knockout.

Read from knockout against wild type: mice without the transgene. The right contrast, Arctic/knockout against Arctic, shows nothing.

Citation — C · Jiménez-Munguía 2021 · S12

The AI wrote: Lbp alone is enough to trigger complement and NF-κB.

The cited paper is about lactoferrin-binding proteins of other bacteria. Here Lbp is the pneumococcal laminin binding protein.

Biology — C · Jiménez-Munguía 2021 · S4

The AI wrote: GBP2 to CASP4 is the route to the IL1B seen with pneumococcus.

Caspase-4 senses cytosolic LPS, which a Gram-positive bacterium does not have.

And one it marked especially good C · Jiménez-Munguía 2021 · S14

IL1B rises only with live pneumococcus, which fits inflammasome activation by its toxin pneumolysin: something the adhesin alone cannot supply. The paper leaves that contrast unexplained.

Flags from one AI reader. Are they right? You and the authors decide.

Your examples



One you marked especially good



One that does not make sense, and why

Your pick, on the presenter laptop.

Thank you

We thank ELIXIR for supporting TELLME.

paintomicsai@gmail.com

Slides and materials

github.com/ConesaLab/SEBiBC2026

All AI sessions: before you leave

elixir

CSIC
Consejo Superior de Investigaciones Científicas

i2sysbio Institute for Integrative Systems Biology

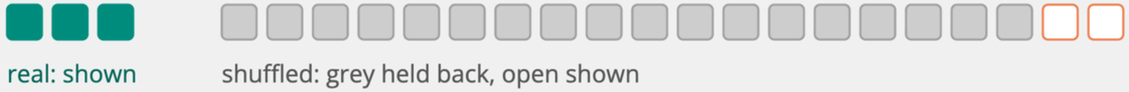
Genomics of Gene Expression Lab

Backup slides

For questions

How we know it works

the checks show all three real network interpretations and hold back 18 of 20 built on shuffled data (Fisher p = 0.006)

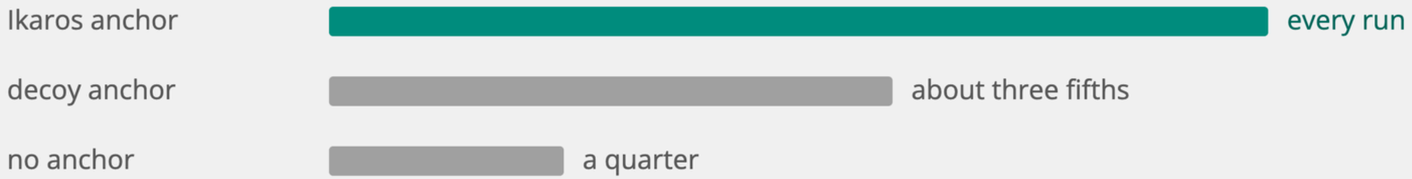


real: shown

shuffled: grey held back, open shown

the direction panel is read correctly in more than 97 % of 164 constructed statements

an Ikaros-anchored walk stays within two steps of Ikaros



Honest misses

the paper's context from MeSH agrees with a careful reader on organism 4 in 5, on cell type and study type about 3 in 5

The walk record

▼ The walk: 66 steps, 10 jumps

module · d=0 from Ikzf1

e1 Pecam1 → Ikzf1

anchor anchor:CollecTRI-human · transcriptional regulation + · against the arrow

Ikzf1's Gene expression shows a strong, sustained upregulation of Ikaros itself (0h +5.19 rising to 6h +6.26, holding ~+6 through 24h), with Proteomics agreeing at most points (0 +2.89, 2 +4.70, 18 +4.75, 24 +4.83) though dipping at 6h/12h (−0.25/−0.24), and DNase-seq open throughout (0h +2.26 to 24h +1.81) — the perturbation is present and sustained from baseline.

e2 Ikzf1 → Inpp5d

anchor anchor:CollecTRI-human · transcriptional regulation

Inpp5d's Proteomics shows a late, progressive loss of SHIP1 protein (2h +0.67 then 12h −1.15, 18h −1.00, 24h −1.32) with DNase-seq closing in parallel (2h −0.61 to 24h −0.84) while Gene expression declines more modestly (6h −0.73, 12h −0.96) — the SHIP1 locus is being shut down at chromatin and protein level from 6–12h onward.

e3 Inpp5d → Syk

OmniPath BcellreceptorBCR · inhibition −

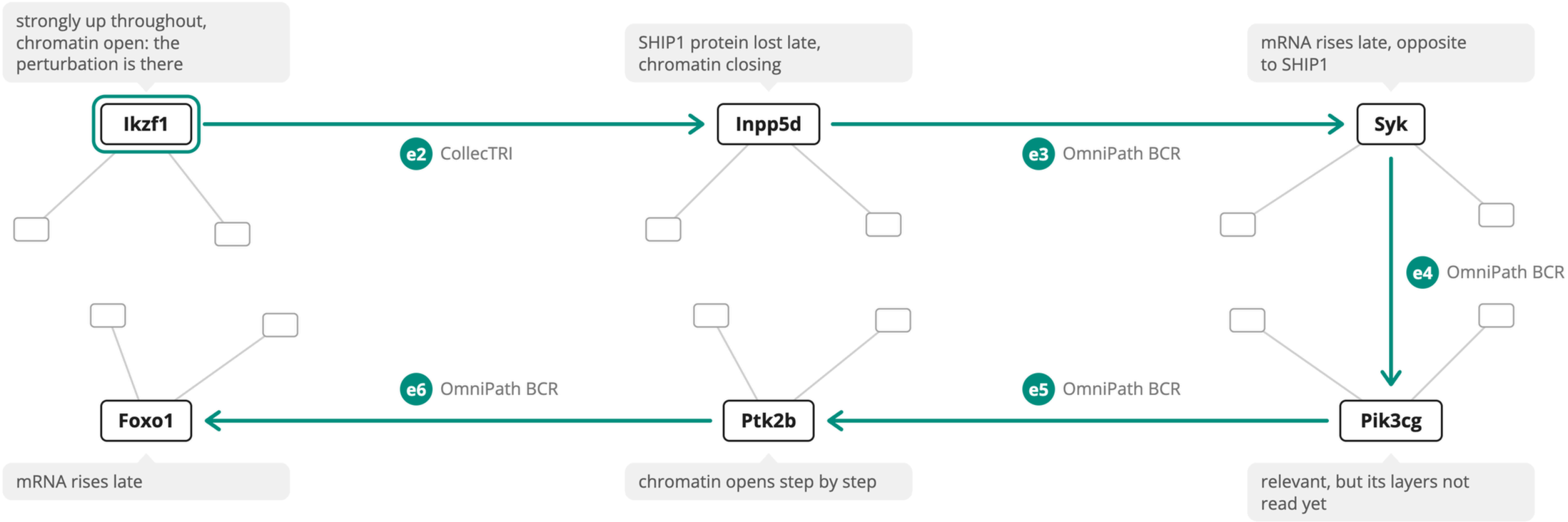
Syk's Gene expression shows a late, strong rise (0h −0.42, 6h +0.11, 12h +1.29, 18h +2.07, 24h +2.33) with DNase-seq open throughout (0h +0.78 to 24h +0.42) while Proteomics is noisy (12h −5.99) — Syk mRNA is induced from 12h onward, opposite in direction to SHIP1.

e4 Syk → Pik3cg

OmniPath BcellreceptorBCR · stimulation +

Pik3cg is a relevant r=0 node (heat 3.76, 57 of 115 neighbours relevant) one step from Syk, d=2 from the anchor — the PI3Kγ catalytic subunit, a direct neighbour of Syk in the BCR/chemokine signalling network; its own layers are not yet read.

One chain, leg by leg



seed (here the anchor) → walked leg ▭ not walked

The six papers

Genomics of Gene Expression Lab

	Paper	What was measured	How the paper used PaintOmics	AI text on the job page
A	**Hernandez 2017** Mol Neurodegener	Mouse · microglia · Arctic model × C5ar1 knockout · 2, 5, 7, 10 months **Gene expression (RNA-seq time course)**	PaintOmics 3: KEGG enrichment of each of nine time-course gene clusters (Additional file 7), used to choose the genes of Figs. 5 and 6.	shown
B	**Wang 2019** iScience	Human · stem-cell-derived hepatic progenitors (day 9) · HNF4A DNA-binding-domain deletion and SUMO-site mutant vs wild type **Gene expression + proteomics**	PaintOmics 3: joint transcriptome + proteome KEGG enrichment of each mutant (section "Integrating Transcriptomics and Proteomics Datasets", Fig. 6A).	shown
C	**Jiménez-Munguía 2021** Sci Rep	Human · brain microvascular endothelial cells · live S. pneumoniae or its adhesion protein Lbp vs untreated cells **Gene expression (RNA-seq)**	PaintOmics: KEGG pathways and molecular functions of the genes responding to both treatments (Table 4, Fig. 2A, Suppl. dataset 1.8).	shown
D	**Balzano-Nogueira 2021** Genome Biol	Human · TEDDY birth cohort · children who later develop islet autoantibodies vs matched controls · 12, 9, 6, 3 and 0 months before seroconversion **Gene expression + metabolomics (with lipid classes)**	PaintOmics 3: the predictive signature projected onto KEGG pathway maps, from which the metabolic and immune models of Figs. 5 and 6 were drawn.	held back by the graph-artifact check; one click shows it
E	**Rubio 2021** Sci Rep	Human · patients with vs without MHE · blood cells and serum **Gene expression + metabolomics + serum cytokines**	PaintOmics 3: KEGG enrichment of the differential genes and the network of significant pathways (Fig. 2, Table S2).	held back by the graph-artifact check; one click shows it
F	**Fàbregas 2018** Nat Commun	Arabidopsis · roots · BRL3-overexpressing vs wild type, well watered and after 5 days without water **Gene expression + metabolomics**	PaintOmics 3: joint KEGG enrichment of root transcripts and metabolites, run for watered and for drought (Suppl. Fig. 11, Suppl. Data 16 and 17).	held back by the graph-artifact check; one click shows it

Two more papers were run but are not in the practice, because their public data is incomplete: Khurana 2023 and Lopez-Ortiz 2022.